DO NOW — RECALL

Lesson 147 · Firewalls, Proxies and Encryption: Layered

Name: _____ Class: _____ Date: _____

1. What attack floods a service with requests to reduce availability?

2. What is a foreign key used for?

3. Which storage type is volatile?

4. Which layer in TCP/IP handles IP routing?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A denial-of-service attack, potentially distributed (DDoS).

2. Model answer

Referencing a candidate key in another relation.

3. Model answer

RAM.

4. Model answer

The internet layer.

By the end, you can…

• Describe how a firewall enforces traffic rules using addresses, ports or connection state.

• Distinguish a forward proxy from a firewall and explain appropriate filtering or caching uses.

• Choose encryption and additional controls for confidentiality without assuming they stop every attack.

Vocabulary — know these terms

firewall — Rule-based network traffic access control.

proxy server — Server that acts as intermediary for client requests.

TLS — Protocol protecting transport of application data.

defence in depth — Overlapping safeguards against different failures.

NEW LEARNING

Read, annotate and explain

Filtering boundary

Firewall applies rules to traffic entering or leaving a network.

Mediating requests

Forward proxy can request resources on behalf of clients, cache and log access.

Protecting content

TLS encrypts data in transit; authentication and patching address other risks.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A school wants to limit outgoing web browsing while protecting pupils' login details over public Wi-Fi. Select appropriate controls.

Client web request

Identify the facts and make an initial prediction.

Proxy and firewall controls

Explain the mechanism or trace the intermediate stage.

HTTPS encrypted transfer

A managed web proxy can filter/log permitted requests; a firewall can restrict traffic flows; HTTPS/TLS encrypts the connection so local eavesdroppers cannot read credentials. Policy, MFA and endpoint protection address further threats.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

State functions of firewall, forward proxy and TLS, one network example each. [6]

Question 2

Explain why encrypted traffic is not the same as authenticated trustworthy content. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Recommend a layered defence for a sixth-form network with remote learners, justifying four different controls. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A forward proxy and a firewall necessarily perform identical tasks.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.